MILITARY SPECIFICATION

SODIUM BIPHOSPHATE AND SODIUM PHOSPHATE SOLUTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a disposable enema unit containing Sodium Biphosphate and Sodium Phosphate solution. (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals. (See 6.4).

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

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AMERICAN PHARMACEUTICAL ASSOCIATION PUBLICATION

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE. FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U.S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a clear, colorless, aqueous solution containing amounts equivalent to Sodium Biphosphate, Monobasic ($NaH_2PO_4.H_2O$), 16.0 percent and Sodium Phosphate, Dibasic, Heptahydrate ($Na_2HPO_4.7H_2O$), 6.0 percent.

3.1.1 Sodium Biphosphate and Sodium Phosphate. If the Sodium Biphosphate, Monobasic, and the Sodium Phosphate, Dibasic, Heptahydrate are of N.F. quality, they may be dissolved in purified water in the manufacture of the solution. As an alternate, the quantity of Sodium Biphosphate may be formed by chemical reaction in the solution.

3.2 Solution.

3.2.1 Assay. The solution shall contain not less than 15.2 percent and not more than 16.8 percent (w/v) of Sodium Biphosphate and not less than 5.7 percent and not more than 6.3 percent (w/v) of Sodium Phosphate, when assayed in 4.4.

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3.2.2 Identification. The enema solution shall meet identification tests for Sodium and Phosphate when tested in accordance with the N.F. Identification-General Tests and the following:

3.2.2.1 Chlorides. 6.5 ml of the enema solution shall not exceed 40 p.p.m. chloride, when tested in accordance with the U.S.P. Chloride Limits Test.

3.2.2.2 Sulfates. 0.5 ml of the enema solution shall not exceed 330 p.p.m. Sulfate, when tested in accordance with the U.S.P. Sulfate Limit Test.

3.2.2.3 Heavy metals. 1.0 ml of the enema solution shall not exceed 10 p.p.m. Heavy Metals, when tested in accordance with Method I of the U.S.P. Heavy Metals Limit Test.

3.2.2.4 Arsenic. 1.0 ml of the enema solution shall not exceed 10 p.p.m. Arsenic (As_2O_3) when tested in accordance with the U.S.P. Arsenic Limit Test.

3.2.2.5 Aluminum, Calcium. A 1:1 dilution of the enema solution shall not become turbid when rendered slightly alkaline to litmus with Ammonia T.S. when viewed downward in a Nessler tube against a white background.

3.2.3 Preservative. The enema solution shall contain suitable preservative(s).

3.2.4 pH. The pH of the solution shall be not less than 5.00 and not more than 5.80, at 25°C., when determined potentiometrically using the U.S.P. method.

3.2.5 Specific gravity. The specific gravity of the finished solution at 25°C./25°C. shall be not less than 1.121 and not more than 1.128.

3.3 Ingredients.

3.3.1 Purified water. The Purified Water used in the manufacture of the solution shall be in accordance with the tests, standards, and requirements of the U.S.P., including any supplements or revisions thereto.

3.3.2 Sodium Biphosphate and Sodium Phosphate. If the Sodium Biphosphate and Sodium Phosphate are not formed by chemical reaction, they shall be in accordance with the tests, standards, and requirements of the N.F., including any supplements or revisions thereto.

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3.3.3 Preservatives. The preservative(s) entering into the finished product shall be of U.S.P. or N.F. quality, or, if not included in either of these compendia, the preservative(s) shall be of the highest pharmaceutical grade.

3.4 Delivery. Not more than 6 months shall have elapsed from the date of manufacture of the product to the date of delivery to the Government.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105, and as specified in Table I.

Table I. Sampling for examination

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled Bottle

4.2.1 For raw material and end item testing. Noncompliance with Table II is cause for rejection.

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Table II. Sampling for raw material and end item testing

Characteristic	Requirement paragraph	Inspection level *	Minimum Sample
End Item:			
Assay	3.2.1		
Identification			
Chloride	3.2.2.1	S-2	
Sulfate	3.2.2.2	(Composite	
Heavy Metals	3.2.2.3	from final	
Arsenic	3.2.2.4	filled	
Aluminum Calcium	3.2.2.5	bottles)	
pH	3.2.4		
Specific gravity	3.2.5		
Ingredients			
Purified Water	3.3.1	**	
Sodium Biphosphate and Sodium Phosphate	3.3.2		

* Where possible, the same sample shall be used for two or more tests.

** Sufficient quantities shall be selected from each lot of each raw material to perform tests indicated in applicable test method. Failure of any sample to comply with test requirements shall be cause for rejection of the lot of material represented.

4.3 Examination. The Sodium Biphosphate and Sodium Phosphate solution shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.2 and 4.2.1.

4.3.1 Classification of defects. Examination will be conducted in accordance with the classification of defects listed in Table III.

Table III. Categories and defects*

Major	
101	Solution is not clear or colorless.
102	Solution not free from foreign matter.
103	Solution has crystal formation.

* Inspection is not restricted to the classified defects listed above.

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4.4 Tests. Assay of solution.

Accurately weigh approximately 28 grams of sample and transfer quantitatively into a 250-ml volumetric flask. Dilute to volume with purified water and mix well.

Pipet 50 ml of the above solution into a 250-ml beaker containing a magnetic stirring bar. Add 15.0 ml of N/1 (0.1N) NaOH volumetric solution followed by 50 ml of purified water, and mix.

Titrate potentiometrically with N/1 hydrochloric acid volumetric solution. Calculate the respective percentages of Sodium Biphosphate and Sodium Phosphate.

As an alternate, a potentiometric recording titrimeter with a calibrated syringe pump that delivers a constant amount of titrant per unit of time may be used. If such an instrument is used, it shall be calibrated prior to running the assays.

Note: The first equivalence point is the Monobasic Sodium Phosphate and the second equivalence point is the Disodium Phosphate.

4.5 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking shall comply wtih the requirements of Section 5. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

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Table IV. Examination of preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 4-1/2 fl. oz. as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate containers. Shall be an unbreakable polyethylene squeeze bottle, which shall be provided with a screwcap closure with a rectal tube which is an integral part of the screwcap. The rectal tube portion exclusive of the screwcap shall have an overall length of not less than 1-7/8 inches nor more than 2-1/8 inches. The rectal tube shall be made of polyethylene, and shall contain a rubber diaphragm valve, shall be prelubricated with White Petrolatum, U.S.P., and shall be enclosed in a suitable polyethylene cap. The immediate container shall be one considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

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5.2.2.2 Unit package. Each unit of issue shall be packaged as specified in 5.2.5 of PPP-C-186. Thickness of board may be 0.012 in lieu of 0.020.

5.2.2.3 Intermediate package. Forty eight (48) units of issue shall be packaged as specified in 5.2.8 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.3.1 Level A. Ninety six (96) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Ninety six (96) units of issue shall be packed as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below.

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) Label information in accordance with commercial practice.

5.4.1.2 Unit package. Each unit package label shall bear the following:

(a) Labeling information in accordance with commercial practice.

(b) the national stock number.

(c) the date of manufacture.

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NOTES:

(1) The national stock number shall not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. The date of manufacture shall be shown in lieu of date packed. Marking shall include the legend:

"DO NOT FREEZE"

5.4.5 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. The date of manufacture shall be shown in lieu of date packed. Marking shall include the legend:

"DO NOT FREEZE"

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) the National Stock No.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-00-619-8915	SODIUM BIPHOSPHATE AND SODIUM PHOSPHATE SOLUTION, Disposable Enema Unit, 4-1/2 fl oz (133 cc)

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

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6.4 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superceded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1323

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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MIL-S-37137

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a. GIVE PARAGRAPH NUMBER AND WORDING

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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